Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/21/2007
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Project Title:	Somatosensory Suppression and Prevention of Post-Flight Re-Entry Disturbances of Posture and Locomotion		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Our overall goal is to develop procedures for alleviating post-flight disturbances of posture and locomotion experienced by astronauts. We propose to continue a program of research which has already shown that either non-mechanically supportive contact of the hand with an environmentally fixed surface or pressure cues from a mobile, hand-carried gyroscope stabilizes postural control in a normal 1g environment, a rotating environment, and parabolic flight. Our strategy will be to improve the effectiveness of the portable haptic device and to extend it to situations involving walking. Our original plan was to investigate three avenues of potential improvement: 1) miniaturizing the sensory gyroscope, 2) comparing the gyroscope to a factor vest, and 3) providing subjects training with both devices. At the beginning of this project, we were directed to accelerate aspects of our work which would lead to the development/definition of countermeasures and would enhance collaborative team interactions with other NSBRI balance prosthesis projects. As a result, we have spent the first year of the project focusing on research to evaluate the

Task Description:	tactile sensory encoding and control of body sway. We have examined how the latency and pattern of postural stabilization are affected by touch contact of the hand with an environmentally fixed surface and whether position or velocity cues about body motion are keys to establishing stabilization. Our results show that natural finger contact forces encode body sway velocity, and that the sensed finger forces in turn control the position of the center of foot pressure. The in-phase oscillation of light finger force in response to body sway velocity is established within 125 ms of initial finger contact; the reduction of the displacement of center of foot pressure by phase-advanced finger forces emerges with a time constant of 1.6 s. These results specify requirements for the motion sensing and stimulus delivery components of a mobile, synthetic system for tactile alleviation of post-flight disturbances of posture and locomotion. Currently, we are evaluating the natural coding and control of body sway by contact of body parts other than the hands with a fixed surface. Our Year 2 goal will be to evaluate whether locomotion can be stabilized by a tactor vest device or a few tactors attached to the hand or other body part delivering synthetic tactile cues that provide naturally timed position and/or velocity cues. This will be examined in a normal 1 g environment as well as in parabolic flight. The results of these studies will provide the basis for designing apparatus and procedures for suppressing re-entry disturbances after spaceflight and hastening sensory- motor readaptation to 1g. This research is most relevant to Risk #14, “Impaired Sensory-Motor Capability to Perform Operational Tasks After Landing and Throughout Re-Adaptation” and Risk #13 “Impaired Sensory-Motor Capability to Perform Operational Tasks During Flight, Entry, and Landing” in the NASA Bioastronautics Roadmap. Continuation of this project will advance the research from the Countermeasure Readiness Level of 4 to a level of 5-6 and will result in a Technology Readiness Level of 3-4.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The haptic cueing theory and technology which we develop in this project can potentially serve as mobility aids to people with neuromotor gait/balance disorders and as a spatial disorientation aid for pilots. We currently have AFOSR funding to develop pilot aids, and we have applied for NIH funding for clinical applications.
Task Progress:	Experiment 1. We measured the latency of postural stabilization after blindfolded subjects (n=12) made fingertip contact with a stable surface. The subjects were tested in a heel-to-toe stance in 25 s duration trials. The test subject stood with right index finger above but not touching a laterally placed force plate. At 12 s into the trial an auditory signal cued the subject to initiate light fingertip contact with the surface. Within 125 ms of initial fingertip contact, the forces at the finger stabilized at ~ 40 g. The amplitude of center of foot pressure excursion decreased more slowly, with a time constant of decay of 1.6 s. The stereotypic pattern of small force fluctuations at the fingertip leading correlated changes in center of foot pressure by several hundred millisceonds was fully evident within 500 ms after fingertip contact. This means postural stabilization begins to occur within a few hundred ms of fingertip contact and is virtually complete within 2 s, and the fingertip is stabilized more rapidly than the rest of the body. We also showed that intermittent contact is as effective as continuous contact in stabilizing sway. This means a mobile haptic aid device need not function contiuously; it could be engaged just in time, as needed. Experiment 2. Twenty one subjects stood in a standard Romberg stance, eyes closed, with and without touch. We computed bivariate cross-correlations among lateral fingertip touch forces (T) and lateral center of foot pressure (CP) position and velocity and body center of mass (CM) position and velocity, as a function of various leads or lags introduced in one variable with respect to the other. The finger force variations were most highly correlated with CP position within a tightly defined temporal relationship in which finger force led CP position by ≈150ms. Finger force was less correlated with CM position, with a highly variable temporal lead. Finger force was highly correlated with CM velocity and lagged CM velocity by a small (≈10ms) but consistent amount. Finger force was significantly less correlated with CP velocity with a more variable temporal relationship. These results suggest that CM sway velocity causes the finger shear forces, which stimulate the cutaneous rapidly adapting mechanoreceptors, and that the sensed finger forces in turn initiate compensatory changes in CP position. The results indicate that encoding body velocity may be the optimal way to employ a haptic stimulator in a mobile postural aid system.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
Abstracts for Journals and Proceedings	Di Zio P, Bakshi A, Lackner JR. "Task-dependent adaptation of posture to Coriolis, cross-coupled stimulation." Society for Neuroscience Annual Meeintg, Washington, DC, July 2006. Society for Neuroscience Annual Meeting, Abstract Book, July 2006. , Jul-2006
Articles in Peer-reviewed Journals	Lackner JR, Di Zio P. "Space motion sickness." Exp Brain Res. 2006 Nov;175(3):377-99. PMID: 17021896 , Nov-2006
Articles in Peer-reviewed Journals	Rabin E, Di Zio P, Lackner JR. "Time course of haptic stabilization of posture." Exp Brain Res. 2006 Mar;170(1):122-6. PMID: 16501964 , Mar-2006